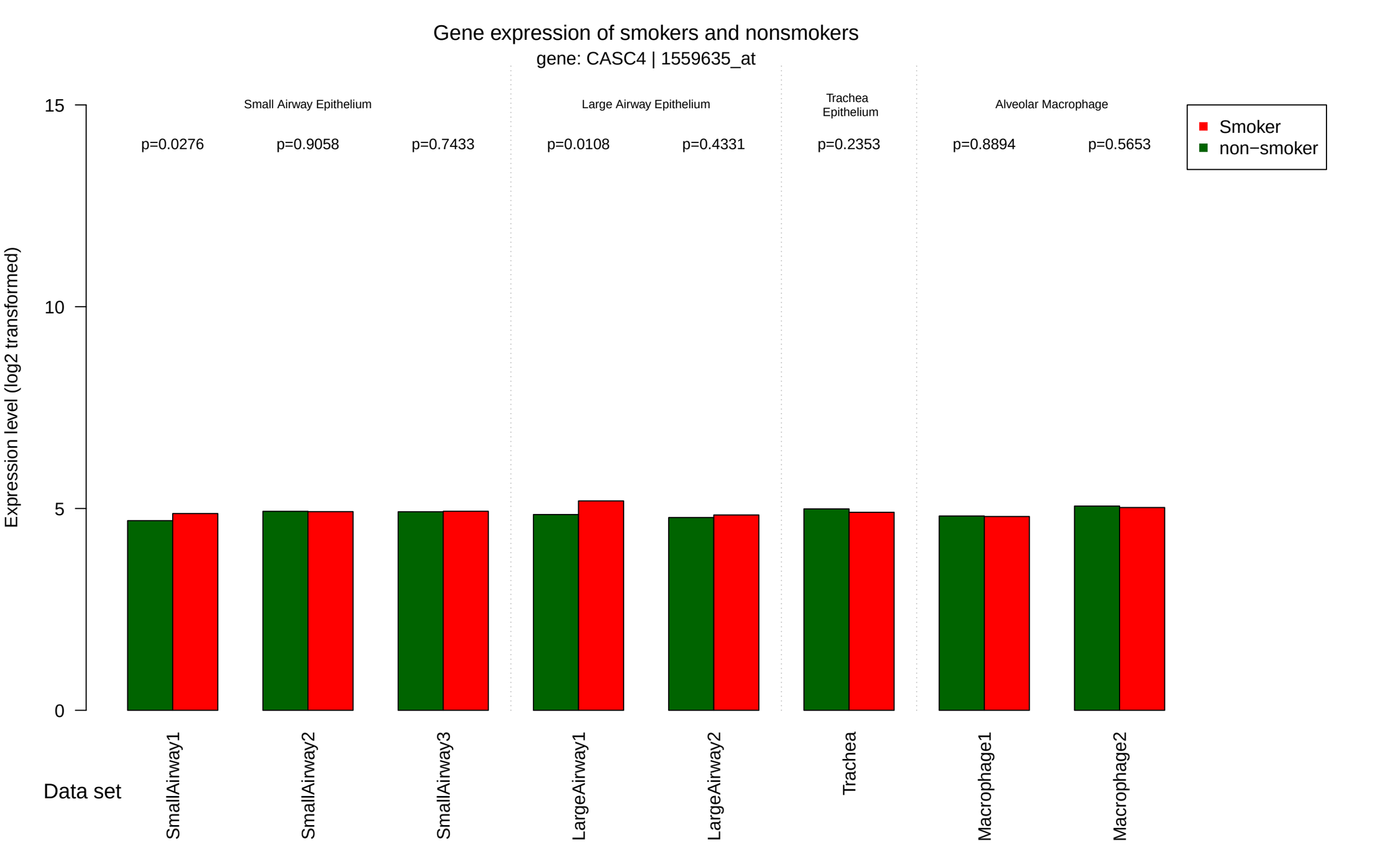
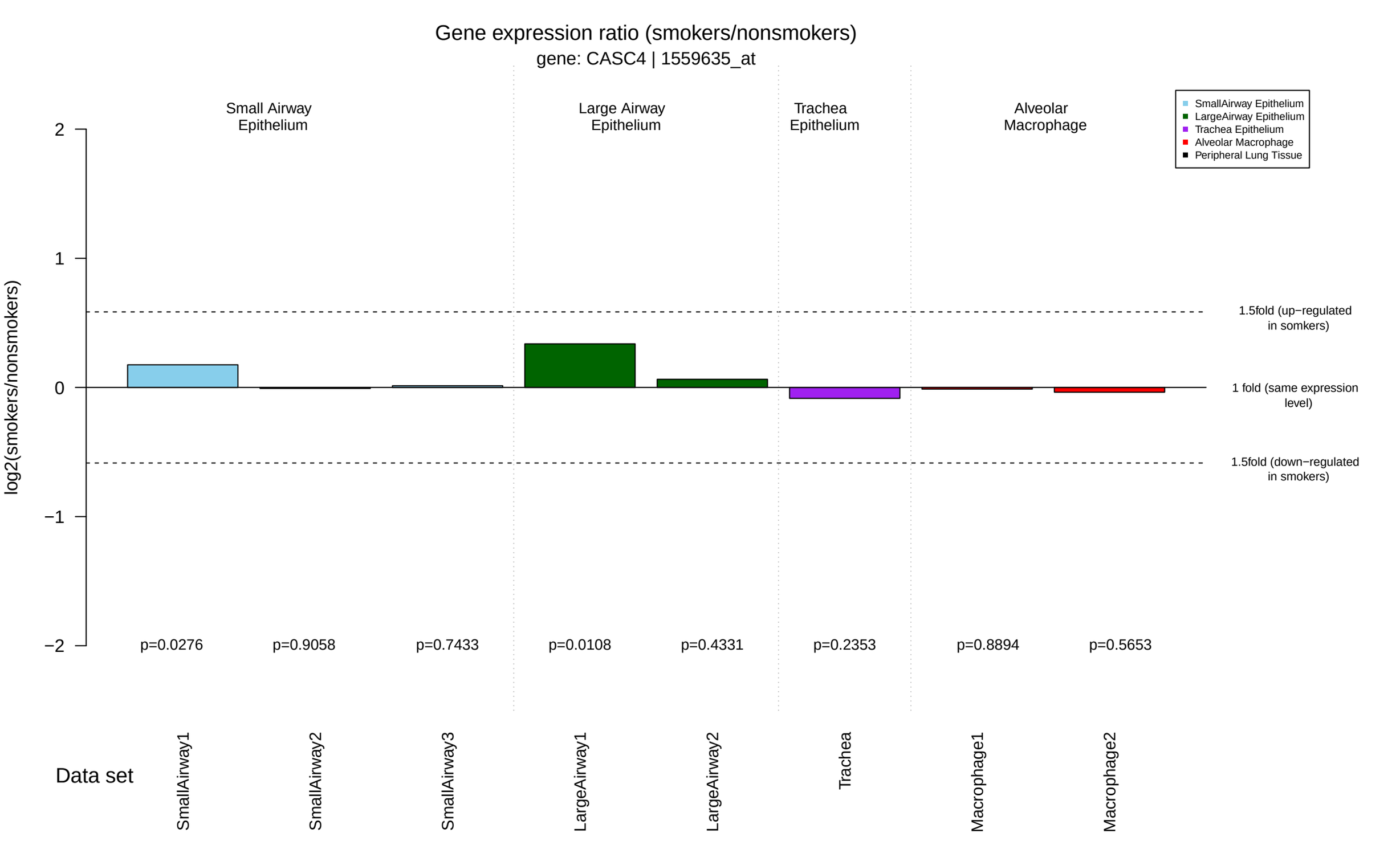
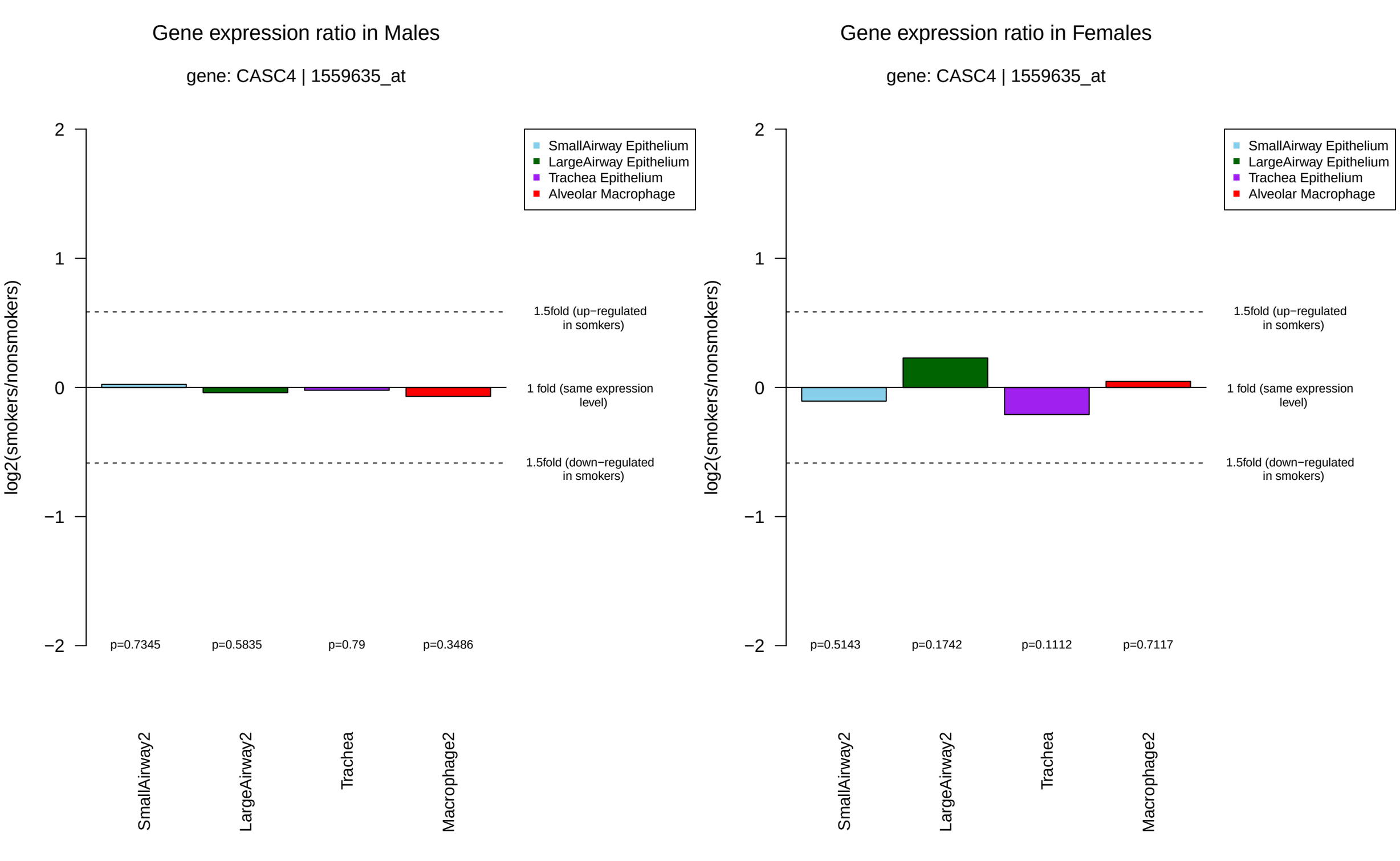
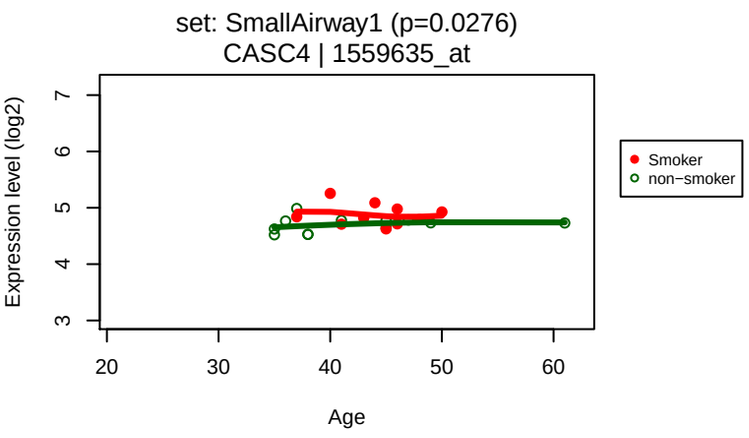
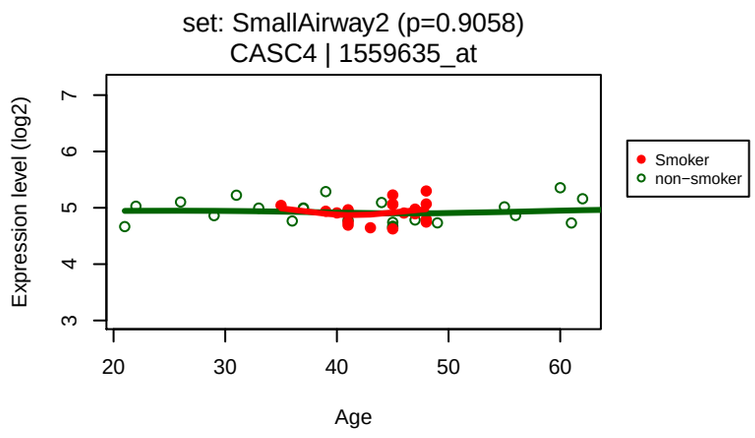
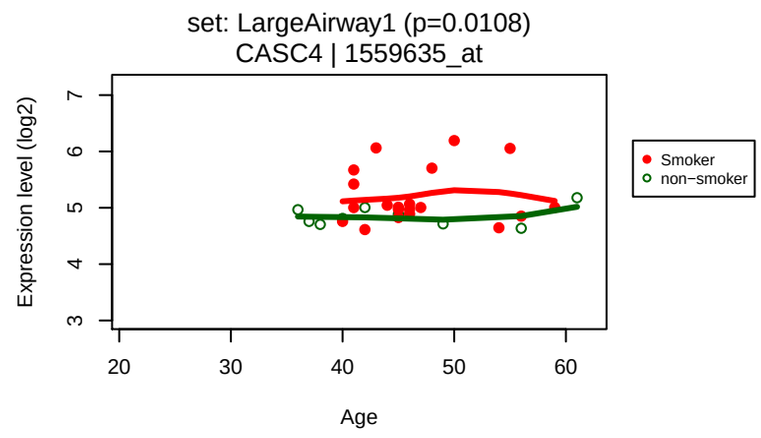

Gene expression of smokers and nonsmokers
gene: CASC4 | 1559635_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0276
p=0.9058
p=0.7433
p=0.0108
p=0.4331
p=0.2353
p=0.8894
p=0.5653
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

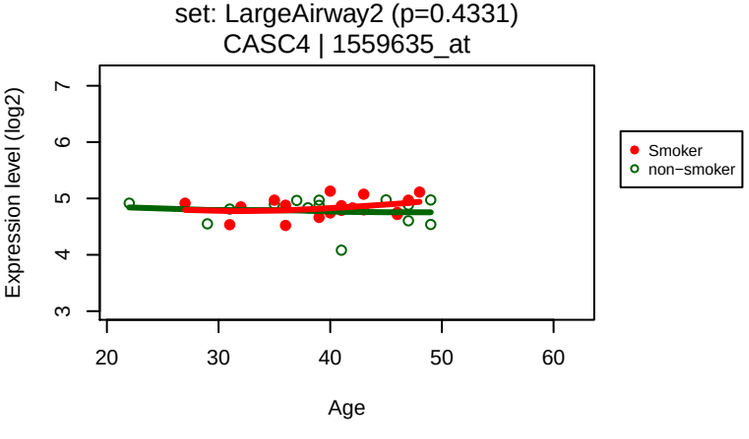

Gene expression ratio (smokers/nonsmokers)
gene: CASC4 | 1559635_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0276
p=0.9058
p=0.7433
p=0.0108
p=0.4331
p=0.2353
p=0.8894
p=0.5653
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

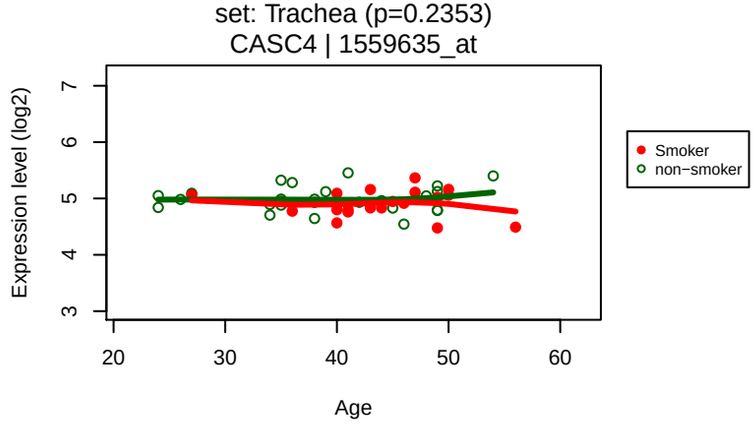

Gene expression ratio in Males
gene: CASC4 | 1559635_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7345
p=0.5835
p=0.79
p=0.3486
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASC4 | 1559635_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5143
p=0.1742
p=0.1112
p=0.7117
SmallAirway2
LargeAirway2
Trachea
Macrophage2

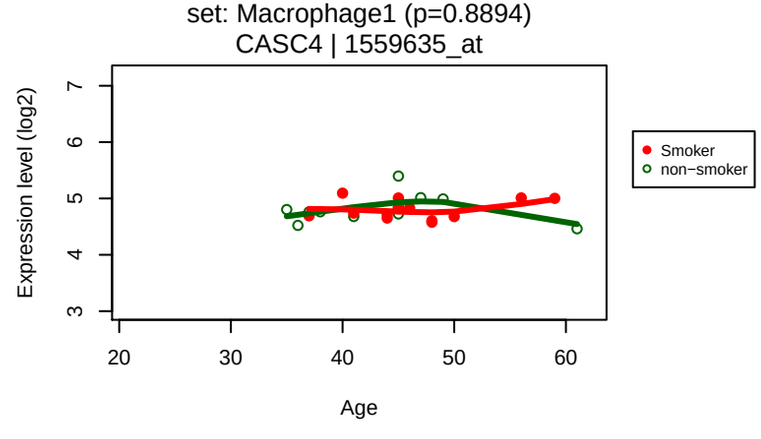

set: SmallAirway1 (p=0.0276)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker

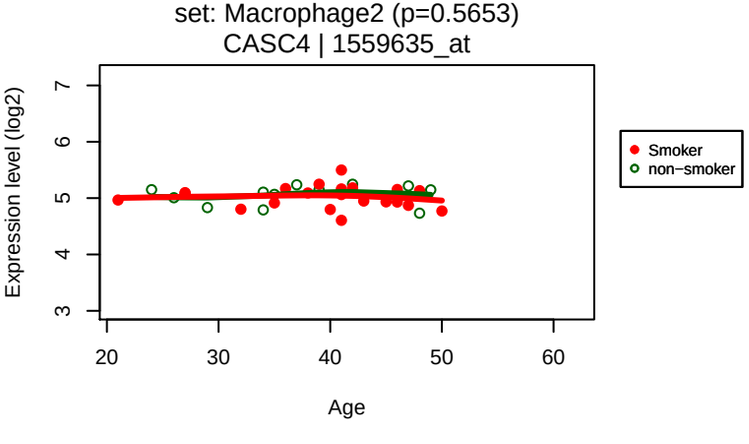

set: SmallAirway2 (p=0.9058)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker

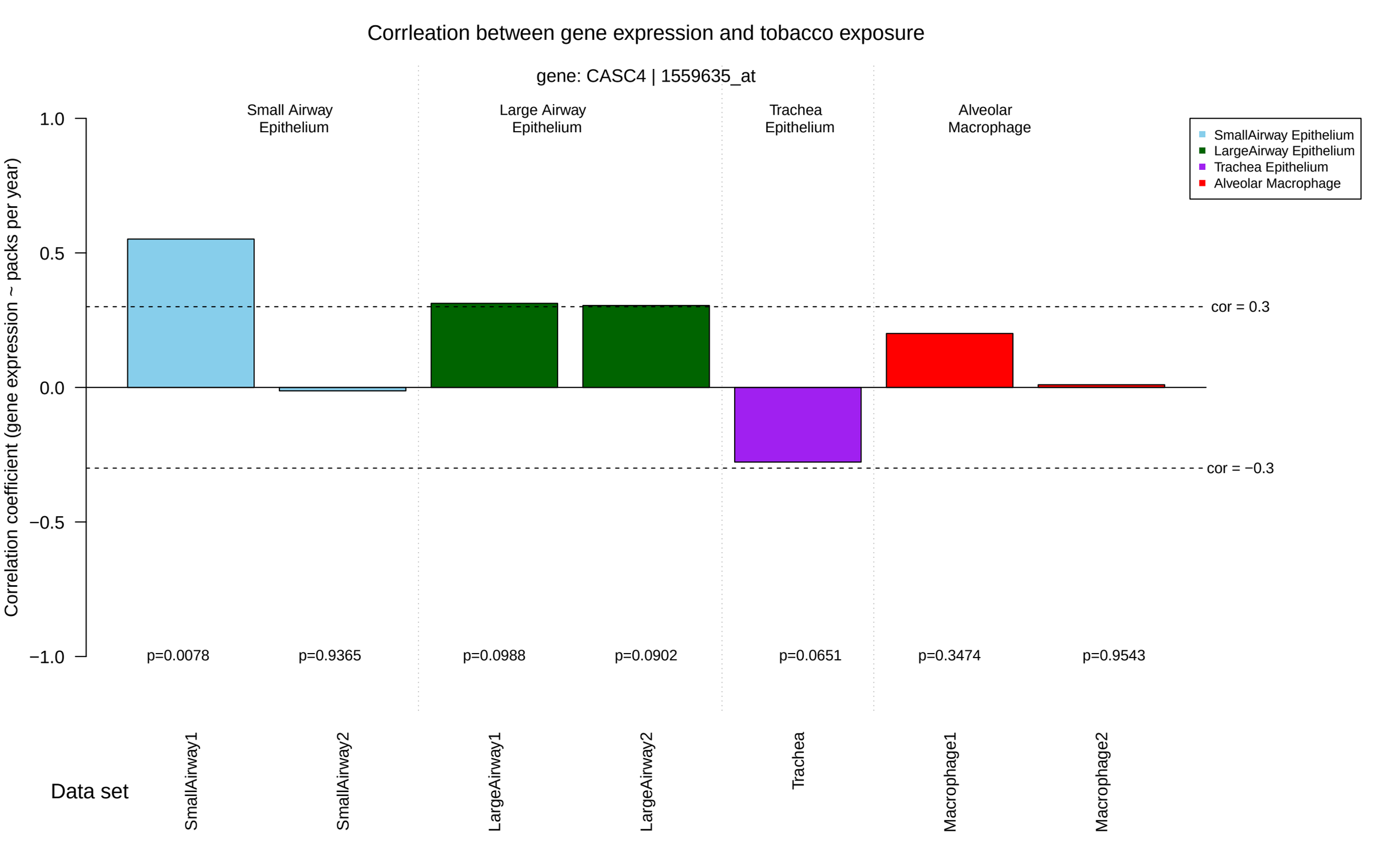

set: LargeAirway1 (p=0.0108)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4331)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2353)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8894)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5653)
CASC4 | 1559635_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASC4 | 1559635_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0078
p=0.9365
p=0.0988
p=0.0902
p=0.0651
p=0.3474
p=0.9543
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2